PRINCIPLES OF MACROECONOMICS

Chapter 16: Inflation, Disinflation, and Deflation

Money, Inflation, and Unemployment in the Short Run and Long Run

ECON_OX 112 · November 12, 2026

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Learning Objectives

- A · Explain seigniorage, the inflation tax, and how people respond to high inflation.
- B · Explain the short-run inflation–unemployment trade-off and how supply shocks change it.
- C · Explain why expectations adjustment leads to a vertical long-run Phillips curve at NAIRU.

16.1 Money and Inflation

- Explain seigniorage, the inflation tax, and how people respond to high inflation.

> Start with money creation: who gains resources, who loses purchasing power, and how do people adapt?

The Classical Model of Money and Prices

Long-run monetary neutrality

Other conditions fixed, increasing the money supply raises the price level rather than real output.

High inflation

Frequent wage and price adjustments make the classical flexible-price approximation useful.

Low, stable inflation

With sticky wages and prices, Keynesian models help explain short-run changes in output and employment.

ECB, Scope of monetary policy

Seigniorage and the Inflation Tax

M: state-issued money stock · ΔM: new money issued · P: price level

Seigniorage

Nominal revenue: $\Delta M = M_t - M_{t-1}$. Real revenue: $\Delta M / P_t$.

The inflation tax

Rising prices reduce the purchasing power of money held by the public.

Government finance

Money issuance can finance spending, while the resulting inflation imposes a cost on money holders.

IMF, Suriname: Toward Stability and Growth (2009), Box 5.2

The Logic of Hyperinflation

Money-financed deficits

The government relies on money creation to cover a persistent financing gap.

Avoiding the inflation tax

As money loses purchasing power, people hold fewer real domestic money balances.

A reinforcing process

If the government issues money faster to obtain the same resources, inflation and avoidance can reinforce each other.

Nachega, IMF Working Paper 05/221 (2005)

Money, Prices, and Wages During Hyperinflation

Function	During hyperinflation
Store of value	People may save in dollars, gold, or goods to limit losses on domestic money.
Unit of account	Prices and contracts may be quoted in a more stable currency or indexed.
Medium of exchange	Sellers may prefer foreign currency or another means of payment.

Prices may change daily or within the day; workers seek frequent wage corrections.

Pay intervals may shorten: monthly → biweekly → weekly → daily → within-day.

IMF, Money: At the Center of Transactions · Bundesbank (2012); Bundestag (2023), Germany's hyperinflation · Bundestag, Hyperinflation in Deutschland 1923

Activity: Surviving High Inflation

Bread basket price · local currency units

Month 1	Month 2	Month 3: now
100	120	144

Your group	Negotiating partner
1 Workers	4 Business Owners
2 Retirees	8 Government
3 Supermarkets	7 Banks
5 Landlords	6 Renters

Read your role

Inflation stays at 20% per month. Use the printed incomes, costs and contract terms; note which amounts are forecasts.

Prepare an offer

Choose two actions, one numerical request and one concession. Explain who else could bear a cost.

Revise, negotiate and report

After new information, revise one term. Record the amount, timing and indexation agreed—or why no agreement was reached.

16.2 Moderate Inflation and Disinflation

- Explain the short-run inflation–unemployment trade-off and how supply shocks change it.

> From protecting purchasing power to a policy dilemma: what happens to jobs when inflation is reduced?

Moderate Inflation and Disinflation

The political business cycle

Elected leaders may seek expansion before elections to raise output and reduce unemployment.

The cost of disinflation

Contractionary monetary or fiscal policy reduces demand and can temporarily raise unemployment.

The political dilemma

Tolerating inflation may postpone the unpopular loss of output and jobs associated with reducing it.

Keller and May (1984), The Presidential Political Business Cycle of 1972 · Federal Reserve Board, Waller (2023), The Unstable Phillips Curve

Potential Output and Unemployment

Potential output, Y^P

The real GDP the economy can sustain with resources used at normal rates.

The output gap

100 × (actual real GDP − potential real GDP) / potential real GDP

The unemployment rate

Actual unemployment = natural unemployment + cyclical unemployment.

Okun’s Law

Output gaps and unemployment deviations from the natural rate move in opposite directions.

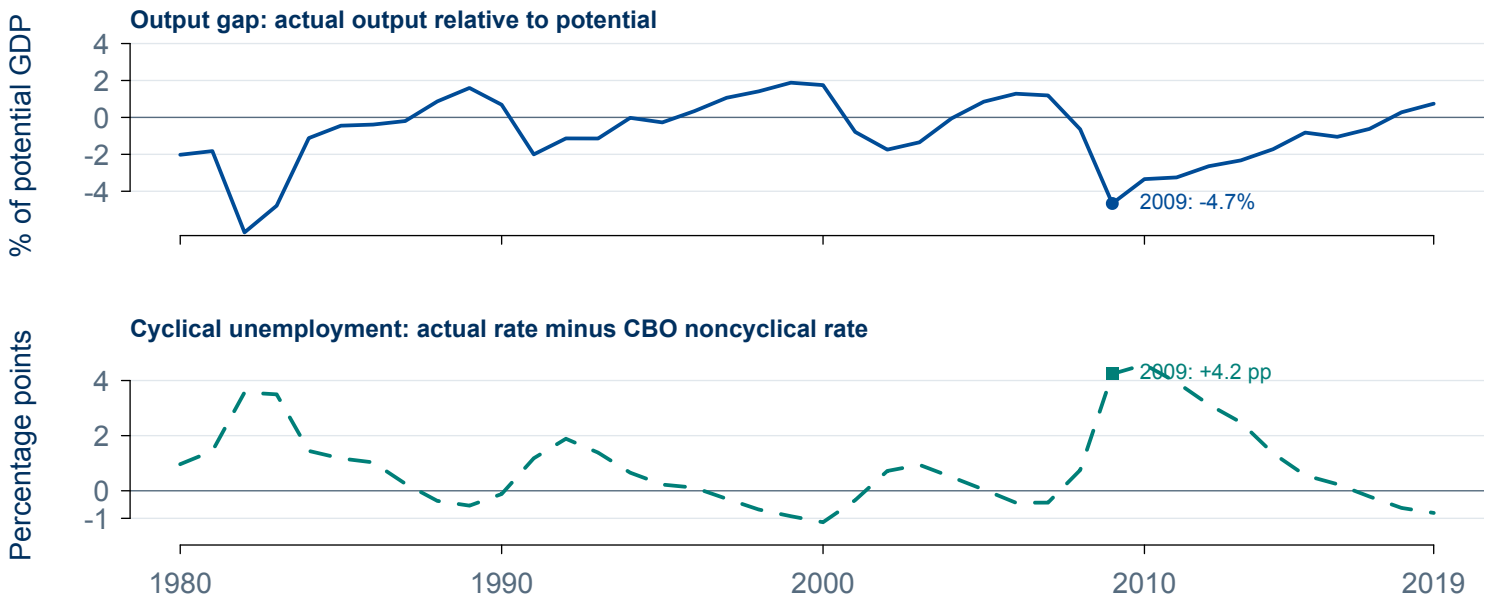
Output	Output gap	Unemployment	Cyclical unemployment
Actual = potential	Zero	Actual = natural	Zero
Actual > potential	Positive	Actual < natural	Negative
Actual < potential	Negative	Actual > natural	Positive

Output gap (%) ≈ −2 × (u − u*)

u*: natural unemployment rate. u − u*: cyclical unemployment, in percentage points. The coefficient is an approximate US rule of thumb.

Federal Reserve Board, Principles for Monetary Policy · Daly and Hobijn, FRBSF (2010), Okun’s Law

Cyclical Unemployment and the Output Gap



Source: CBO, February 2026 historical economic data; annual values, 1980–2019. Original calculations.

Estimated potential output

CBO combines sustainable labor input, capital services, and productivity.

Estimated noncyclical unemployment

CBO’s 2023 approach reweights benchmark demographic unemployment rates as the workforce changes.

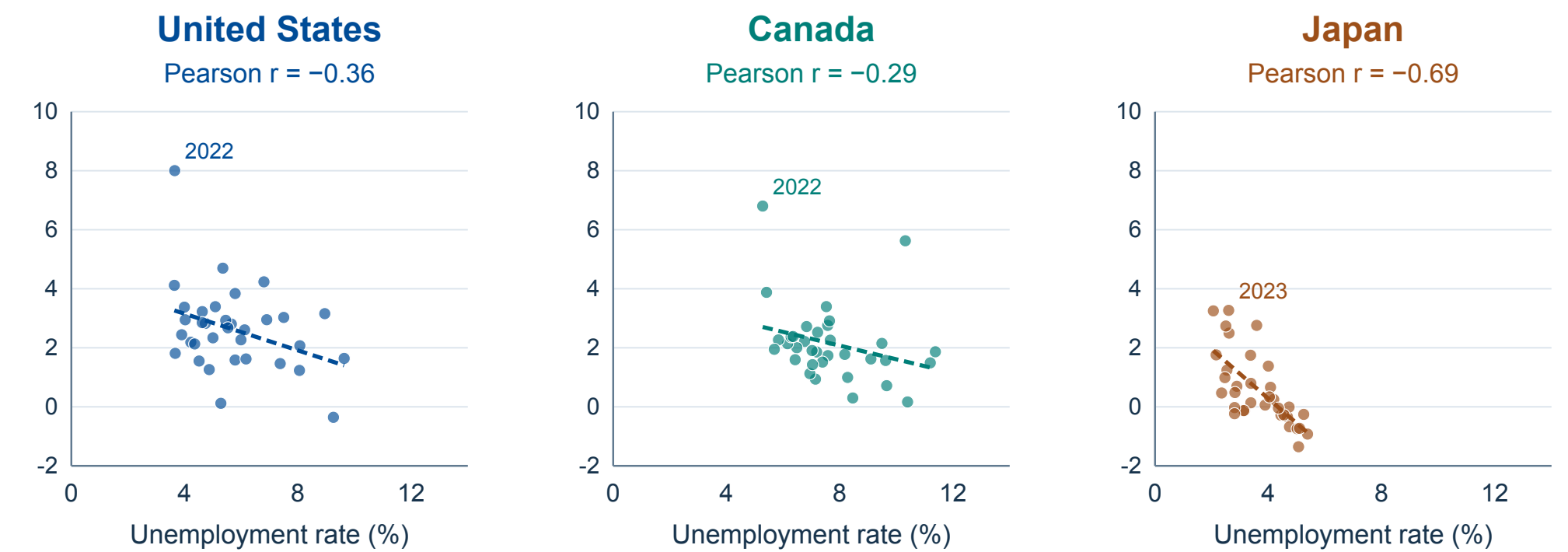
CBO, February 2026 historical economic data · CBO, Approaches to Estimating the Noncyclical Rate (2023) · CBO, Forecasting methods (2018)

Introducing the Phillips Curve: The Data

How are inflation and unemployment related? Each point is one year.

CPI inflation (% per year)

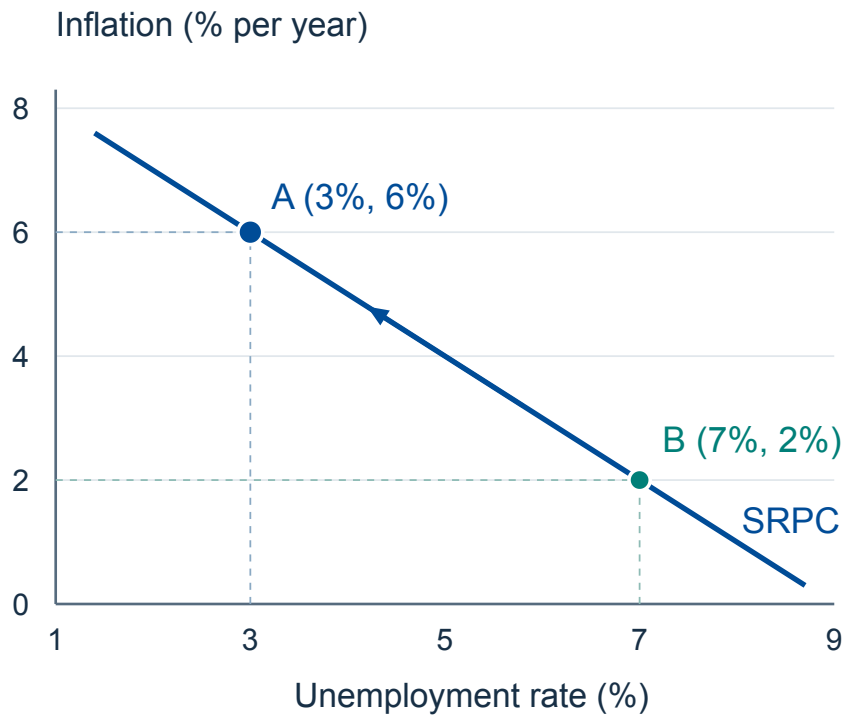
1991–2024 · 34 annual observations per country



World Bank: WDI (IMF/ILO). CC BY 4.0. Original selection, chart and OLS/Pearson calculations.

Same scales. Unemployment: modeled ILO estimates. Country CPI baskets differ. Fits are descriptive, not causal.

The Short-Run Phillips Curve



New Zealand economist A.W.H. Phillips (1958): UK unemployment and nominal wage growth, 1861–1957. Here: consumer-price inflation.

What the curve shows

A downward relationship between inflation and unemployment, with expectations and supply conditions fixed.

A → B: demand restraint

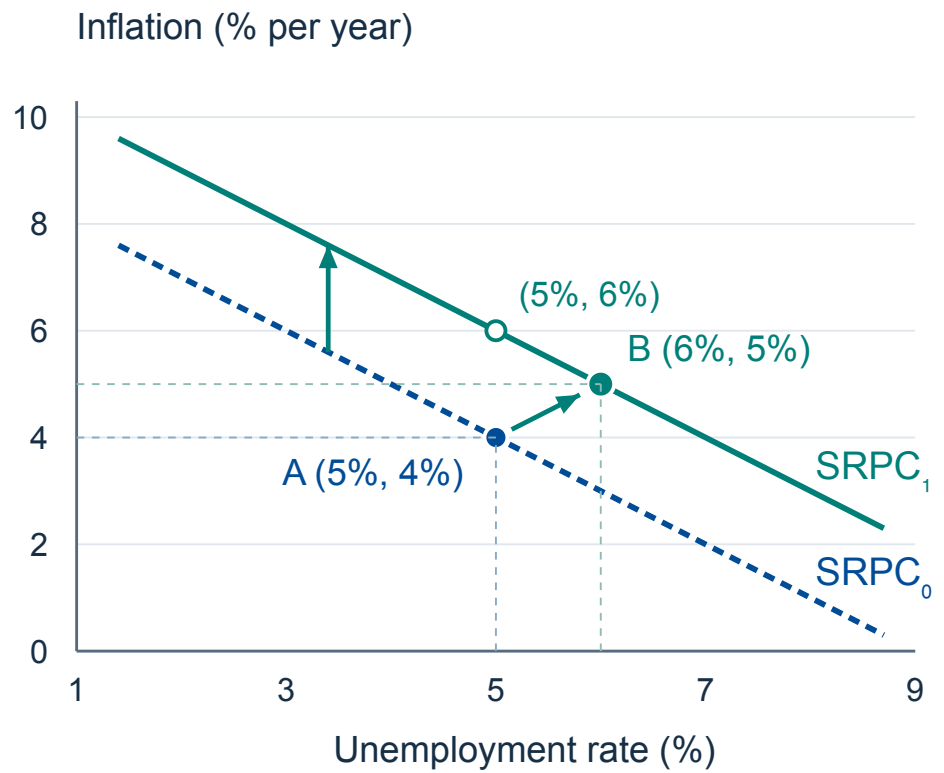
Weaker labor demand limits wage increases. Inflation falls, while unemployment rises.

B → A: demand expansion

A tighter labor market strengthens wage bargaining and adds price pressure; unemployment falls.

Reserve Bank of New Zealand, Monetary policy objectives (2019) · Federal Reserve Board, Waller (2023), The Unstable Phillips Curve

Supply Shocks and the Phillips Curve



A: before an oil shock

Unemployment is 5% and inflation is 4% on $SRPC_0$.

Higher oil costs: an upward shift

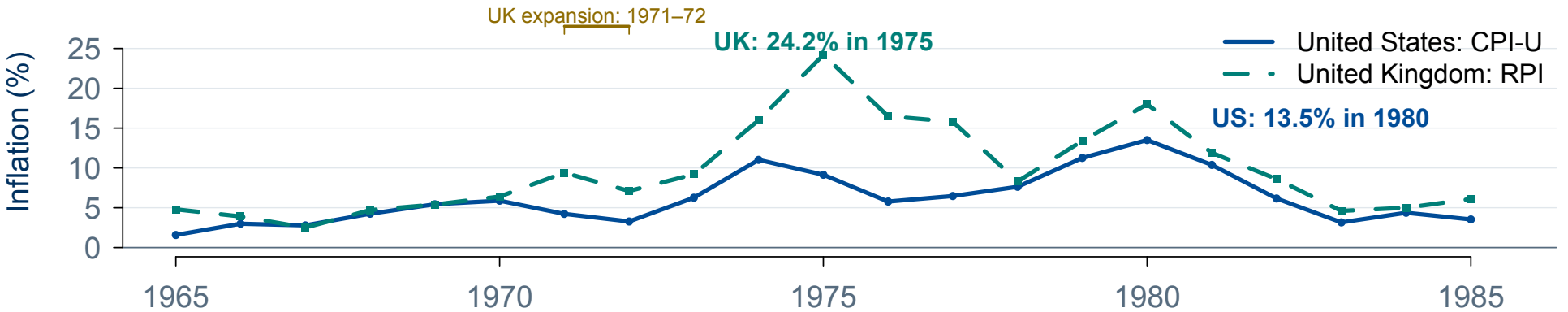
Oil costs raise inflation at each unemployment rate. Falling costs reverse the shift.

B: inflation and unemployment rise

Firms reduce output and hiring. Workers may seek wage increases to restore lost purchasing power.

Federal Reserve Board, Waller (2023), The Unstable Phillips Curve · RBA, Causes of Inflation

Oil Shocks and Stagflation



Annual averages. Source: CBO February 2026 historical CPI-U file (BLS); ONS CZBH, RPI. Different national indexes.

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1973–1975: after the 1973–74 oil embargo

US inflation: 6.3% → 9.1%; unemployment: 4.9% → 8.5%.

1978–1980: the Iranian revolution and second oil shock

US inflation: 7.6% → 13.5%; unemployment: 6.1% → 7.2%.

Stagflation

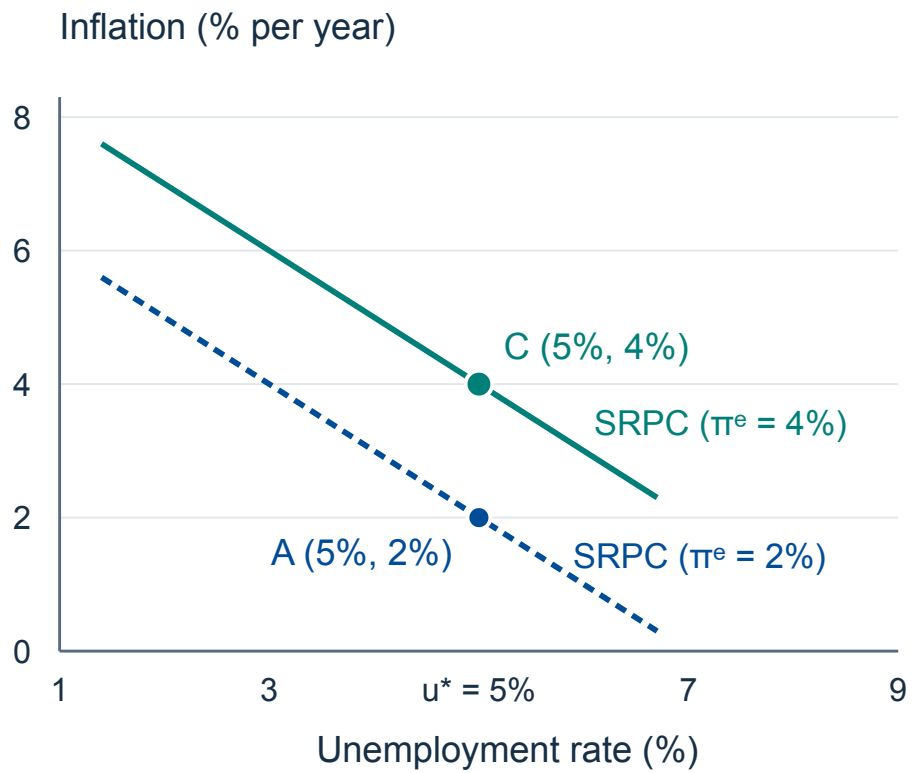
A supply shift can raise inflation and unemployment together while output weakens.

16.3 Inflation and Unemployment in the Long Run

- Explain why expectations adjustment leads to a vertical long-run Phillips curve at NAIRU.

> From the short-run trade-off to its long-run limit: what changes when people come to expect inflation?

Inflation Expectations and the Short-Run Phillips Curve



Expected inflation

The inflation rate employers and workers expect in the near future. Initially, actual and expected inflation are both 2%.

Friedman and Phelps

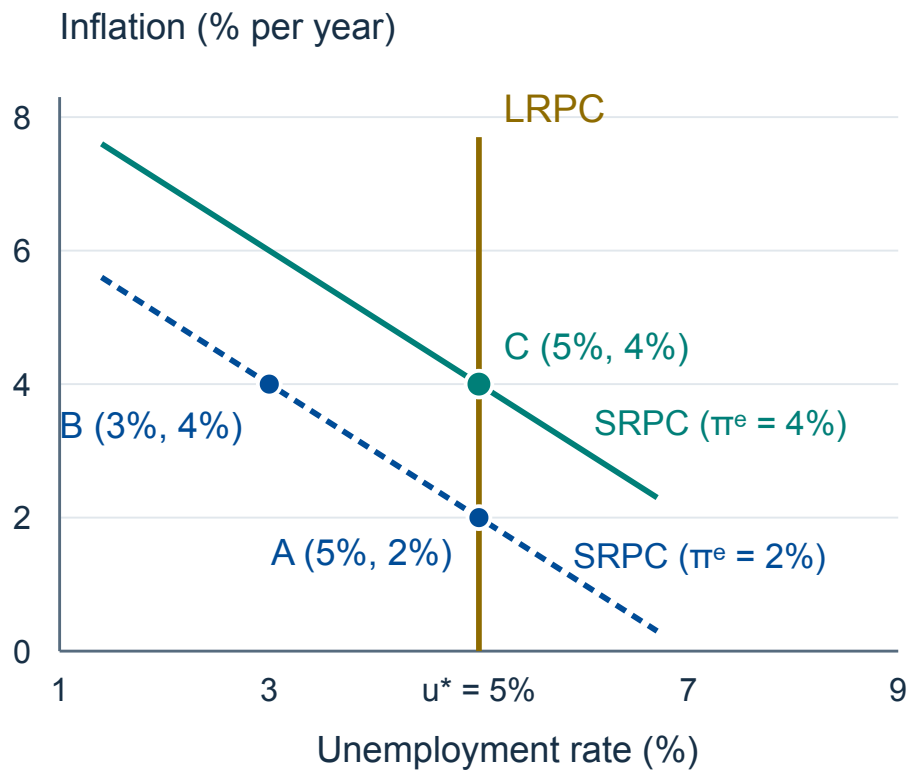
Expectations about future inflation affect wage bargaining and price setting today. Expected inflation rises to 4%.

An upward shift

At 5% unemployment, inflation is now 4% rather than 2%. A higher expected rate shifts SRPC upward.

Federal Reserve Board, Waller (2023), The Unstable Phillips Curve · Lansing, FRBSF (2002), Can the Phillips Curve Help Forecast Inflation?

Adjustment to the Long Run



A: initial long-run equilibrium

Unemployment is 5%; actual and expected inflation are 2%.

B: short-run demand expansion

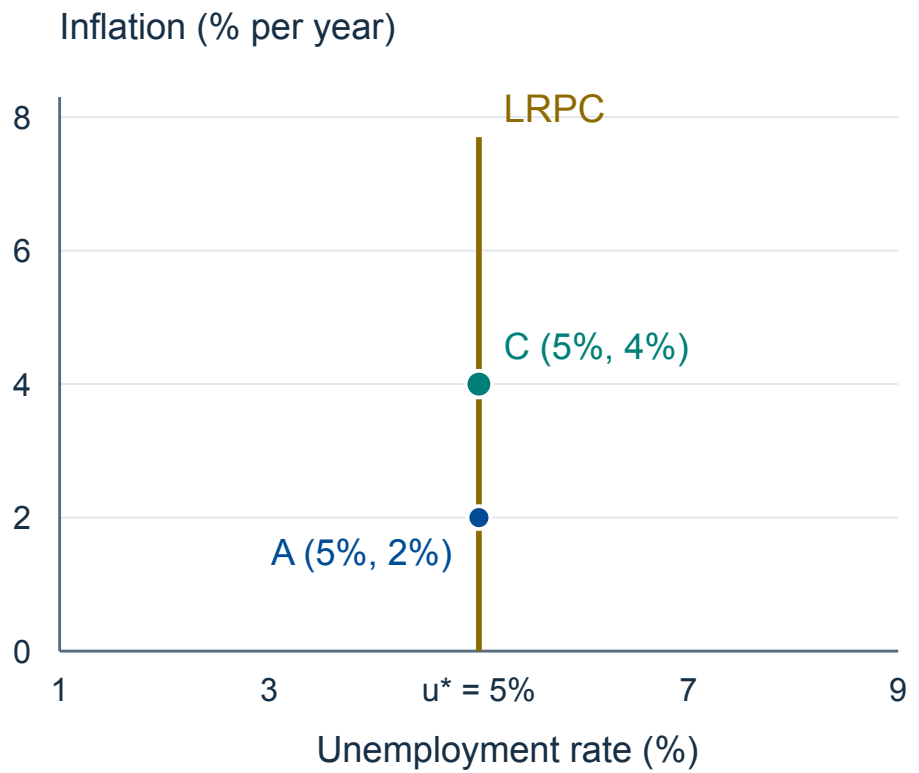
Inflation rises to 4% before expectations adjust. Unemployment temporarily falls to 3%.

C: expectations catch up

Wages and prices incorporate 4% inflation. The short-run curve shifts up; unemployment returns to 5%.

Federal Reserve Board, Waller (2023), The Unstable Phillips Curve · Federal Reserve Board, Mishkin (2007), Estimating Potential Output

The Long-Run Phillips Curve and NAIRU



A vertical long-run curve

Stable inflation of 2% or 4% has the same u* = 5%. Higher fully expected inflation cannot permanently lower unemployment.

NAIRU

Non-Accelerating Inflation Rate of Unemployment: the natural rate in this model, consistent with stable inflation, other conditions fixed.

Inflation pressure

Below u*, inflation tends to rise; above it, to fall. At u*, cyclical unemployment is zero; frictional and structural unemployment remain.

RBA, The Non-Accelerating Inflation Rate of Unemployment · Federal Reserve Board, Mishkin (2007), Estimating Potential Output · Federal Reserve Board, Waller (2023), The Unstable Phillips Curve